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Question 1

- 区别
 - 字节流: 按字节 (8位) 处理数据, 适合处理二进制文件 (如图片, 音频, 视频等); 字节流不对数据进行编码解码, 直接操作字节; 处理非文本数据或需要精确控制字节的内容
 - 字符流: 按字符 (16位) 处理数据, 适合处理文本文件; 字符流会根据编码规则堆字节进行编码和解码, 转化为字符; 处理纯文本数据使用
- 应该使用 `FileInputStream` 流: 因为可以按字节读取文件内容, 非常适合对文件内容进行逐字节操作

Question 2

- 程序输出

```
3
abc
1
dbc
```

- 输出不同的原因: 第一次输入的时候由于字符都被存储为了byte类型, 成功读入了3个, 当第二次尝试读入的时候就只剩下最后一个了, 所以第二次的 `m = 1`, 同理, 第一次tom读完之后为 `abc`, 第二次成功读入一个, tom第一个被覆盖, 所以输出为 `dbc`

Question 3

代码见Question03

The screenshot shows the IntelliJ IDEA IDE with a dark theme. The main editor window displays the code for `Main.java`. The code defines a `Main` class with a `main` method that reads from a file and a `Student` class with private attributes `studentId`, `name`, and `score`, and a constructor. The `main` method also includes a `writeIntoFile` method that writes student data to a file.

```
5 public class Main {
6     public static void main(String[] args) {
27         } catch (IOException e) {
28             System.out.println("failed to read from " + filename);
29         }
30     }
31 }
32
33 class Student { 13 个用法
34     private String studentId; 3 个用法
35     private String name; 3 个用法
36     private double score; 3 个用法
37     // constructor(s) and other methods ...
38     public Student(String studentId, String name, double score) { 3 个用法
39         this.studentId = studentId;
40         this.name = name;
41         this.score = score;
42     }
43
44     @
45     public static void writeIntoFile(Student[] students, String filename) throws IOException {
46         try(BufferedWriter out = new BufferedWriter(new FileWriter(filename))) {
47             for(Student student : students) {
48                 out.write(student.studentId + ", " + student.name + ", " + student.score);
49                 out.newLine();
50             }
51         }
52     }
53 }
```

The bottom panel shows the execution output for the `Main` class:

```
"C:\Program Files\Java\jdk-19\bin\java.exe" "-javaagent:J:\IntelliJ IDEA 2024.1\lib\idea_rt.jar"
written to student.txt
read from student.txt
Student{studentId='2021001', name='John Doe', score=85.5}
Student{studentId='2021002', name='Jane Smith', score=90.0}
```

Question 4

代码见Question04

The screenshot displays the IntelliJ IDEA IDE with a Java file named `Main.java`. The code implements a `main` method that performs three steps: adding data to an array, writing it to a file, and reading it back. The `Student` class is used to represent the data, and `Student.serializeIntoFile` and `Student.deserializeFromFile` methods are used for file operations. The output window shows the execution results, including the file path, the data written to the file, and the data read back from the file.

```
1 import java.io.*;
2
3 public class Main {
4     public static void main(String[] args) {
5         // step1. add data into this array
6         Student[] students = {
7             new Student("2021001", "John Doe", 85.5),
8             new Student("2021002", "Jane Smith", 90.0)
9         };
10        // step2. write this array into file
11        String filename = "student.ser";
12        try {
13            Student.serializeIntoFile(students, filename);
14            System.out.println("written to " + filename);
15        } catch (IOException e) {
16            System.out.println("failed to write to " + filename);
17        }
18        // step3. read students from file and print
19        try {
20            Student[] result = Student.deserializeFromFile(filename);
21            System.out.println("read from " + filename);
22            for(Student student : result) {
23                System.out.println(student);
24            }
25        }
26    }
27 }
```

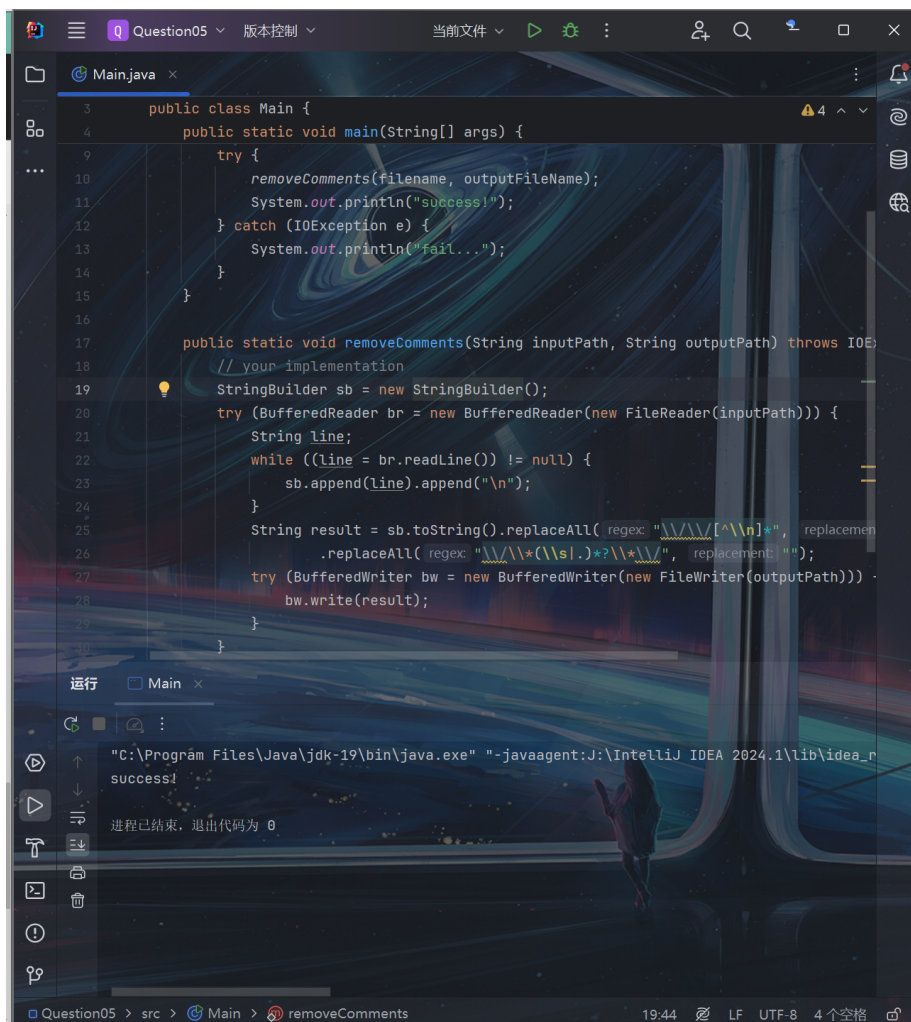
运行 Main

```
"C:\Program Files\Java\jdk-19\bin\java.exe" "-javaagent:J:\IntelliJ IDEA 2024.1\lib\idea_r
written to student.ser
read from student.ser
Student{studentId='2021001', name='John Doe', score=85.5}
Student{studentId='2021002', name='Jane Smith', score=90.0}

进程已结束，退出代码为 0
```

Question 5

代码见Question05



Question 6

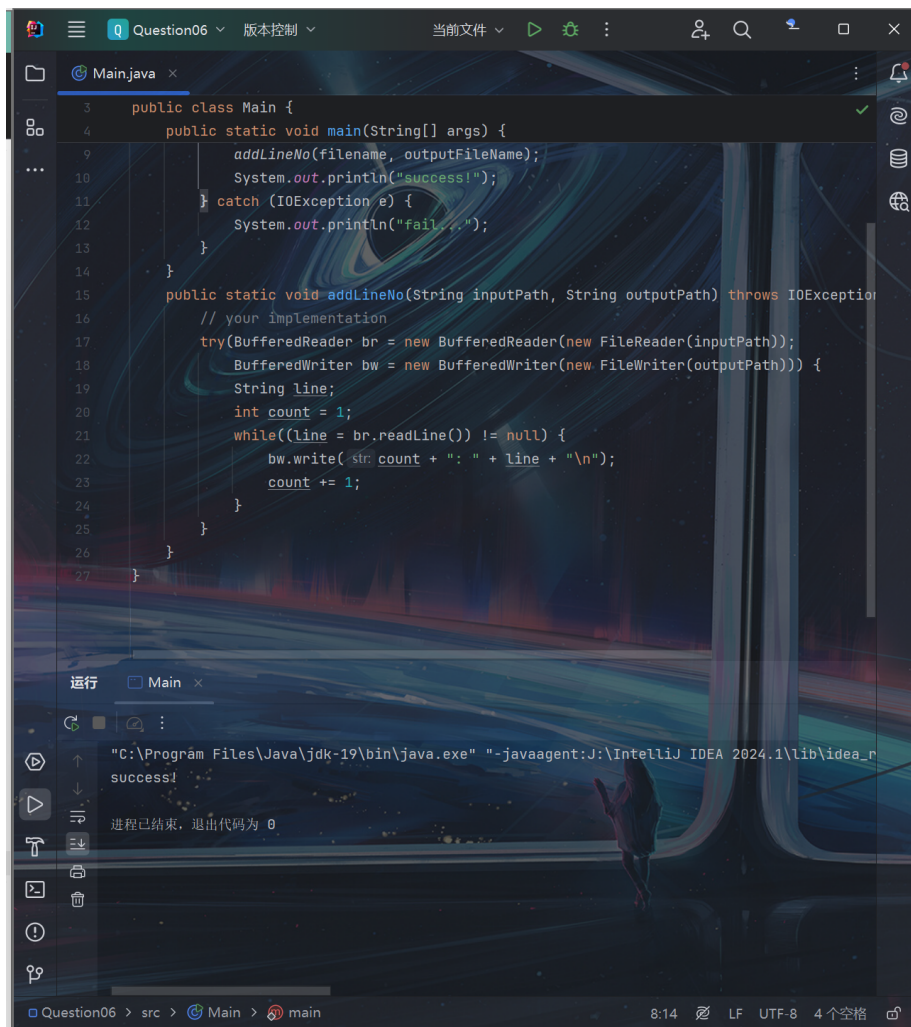
代码见Question06



```
1 sa
2 d
3 sd
4 s
5 d
6 as
7 ds
8 d
9 a
10 s
11 d
```

```
1 1: sa
2 2: d
3 3: sd
4 4: s
5 5: d
6 6: as
7 7: ds
8 8: d
9 9: a
10 10: s
11 11: d
12
```

```
"C:\Program Files\Java\jdk-19\bin\java.exe" "-javaagent:J:\IntelliJ IDEA 2024.1\lib\idea_r
success!"
进程已结束，退出代码为 0
```



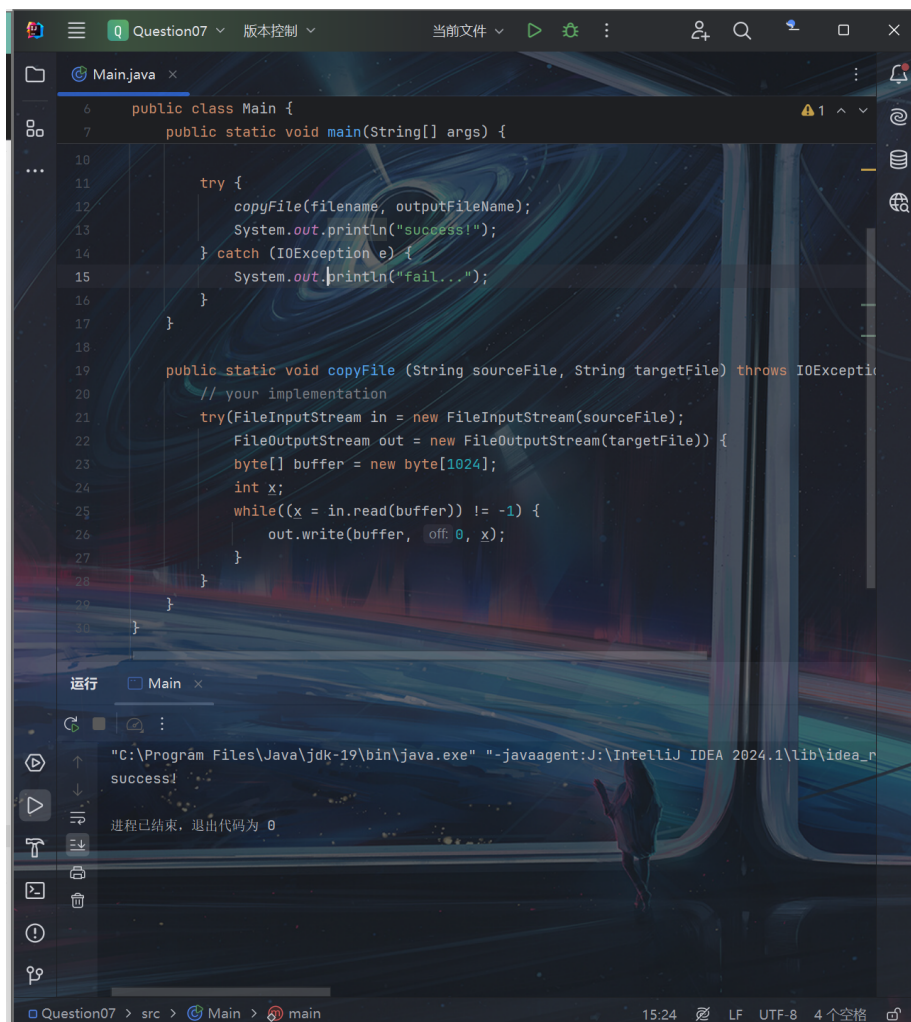
```
3 public class Main {
4     public static void main(String[] args) {
5         addLineNo(filename, outputFileName);
6         System.out.println("success!");
7     } catch (IOException e) {
8         System.out.println("fail...");
9     }
10 }
11
12 public static void addLineNo(String inputPath, String outputPath) throws IOException {
13     // your implementation
14     try(BufferedReader br = new BufferedReader(new FileReader(inputPath));
15         BufferedWriter bw = new BufferedWriter(new FileWriter(outputPath))) {
16         String line;
17         int count = 1;
18         while((line = br.readLine()) != null) {
19             bw.write(str: count + ": " + line + "\n");
20             count += 1;
21         }
22     }
23 }
24 }
```

运行 Main

```
"C:\Program Files\Java\jdk-19\bin\java.exe" "-javaagent:J:\IntelliJ IDEA 2024.1\lib\idea_r
success!
进程已结束，退出代码为 0
```

Question 7

代码见Question07



Question 8

代码见Question08

```
Windows PowerShell
版权所有 (C) Microsoft Corporation. 保留所有权利。

安装最新的 PowerShell, 了解新功能和改进! https://aka.ms/PSWindows

加载个人及系统配置文件用了 3180 毫秒。
(base) PS C:\Users\12298\Desktop\2024_Fall\Java\Lab7\Question08> tree YourNoteDir
卷 OS 的文件夹 PATH 列表
卷序列号为 72D3-787D
C:\USERS\12298\DESKTOP\2024_FALL\JAVA\LAB7\QUESTION08\YOURNOTEDIR
├── 离散数学
├── 计算机硬件基础
├── 马克思主义基本原理
└── 马原的马原

(base) PS C:\Users\12298\Desktop\2024_Fall\Java\Lab7\Question08> tree MyNoteDir
卷 OS 的文件夹 PATH 列表
卷序列号为 72D3-787D
C:\USERS\12298\DESKTOP\2024_FALL\JAVA\LAB7\QUESTION08\MYNOTEDIR
├── 离散数学
├── 计算机硬件基础
├── 马克思主义基本原理
└── 马原的马原

(base) PS C:\Users\12298\Desktop\2024_Fall\Java\Lab7\Question08> |
```

```
Question08 版本控制 当前文件
Main.java
6 public class Main {
19 public static void copyDirectory(String sourceDir, String targetDir) throws IOException {
21     File sourceFolder = new File(sourceDir);
22     File targetFolder = new File(targetDir);
23
24     if(!targetFolder.exists()) {
25         targetFolder.mkdirs();
26     }
27
28     File[] files = sourceFolder.listFiles();
29     for(File file : files) {
30         if(file.isDirectory()) {
31             copyDirectory(file.getAbsolutePath(), targetDir, targetFolder.getAbsolutePath());
32         } else {
33             copyFile(file.getAbsolutePath(), targetFile: targetFolder.getAbsolutePath());
34         }
35     }
36 }
37
38 public static void copyFile (String sourceFile, String targetFile) throws IOException {
39     // your implementation
40     try(FileInputStream in = new FileInputStream(sourceFile);
41         FileOutputStream out = new FileOutputStream(targetFile)) {
42         byte[] buffer = new byte[1024];
43     }
44 }
45 }

运行 Main
"C:\Program Files\Java\jdk-19\bin\java.exe" "-javaagent:J:\IntelliJ IDEA 2024.1\lib\idea_r
success!
进程已结束, 退出代码为 0

Question08 > src > Main > copyDirectory 25:35 LF UTF-8 4个空格
```